



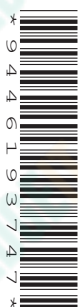
Cambridge International AS & A Level

CANDIDATE
NAME
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**PHYSICS****9702/34**

Paper 3 Advanced Practical Skills 2

May/June 2024**2 hours**

You must answer on the question paper.

You will need: The materials and apparatus listed in the confidential instructions

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You will be allowed to work with the apparatus for a maximum of 1 hour for each question.
- You should record all your observations in the spaces provided in the question paper as soon as these observations are made.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 40.
- The number of marks for each question or part question is shown in brackets [].

For Examiner's Use	
1	
2	
Total	

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You may not need to use all of the materials provided.

1 In this experiment, you will investigate the equilibrium position of a pulley system.

- (a) • Assemble the apparatus as shown in Fig. 1.1 with the rods of the stands approximately 50 cm apart.

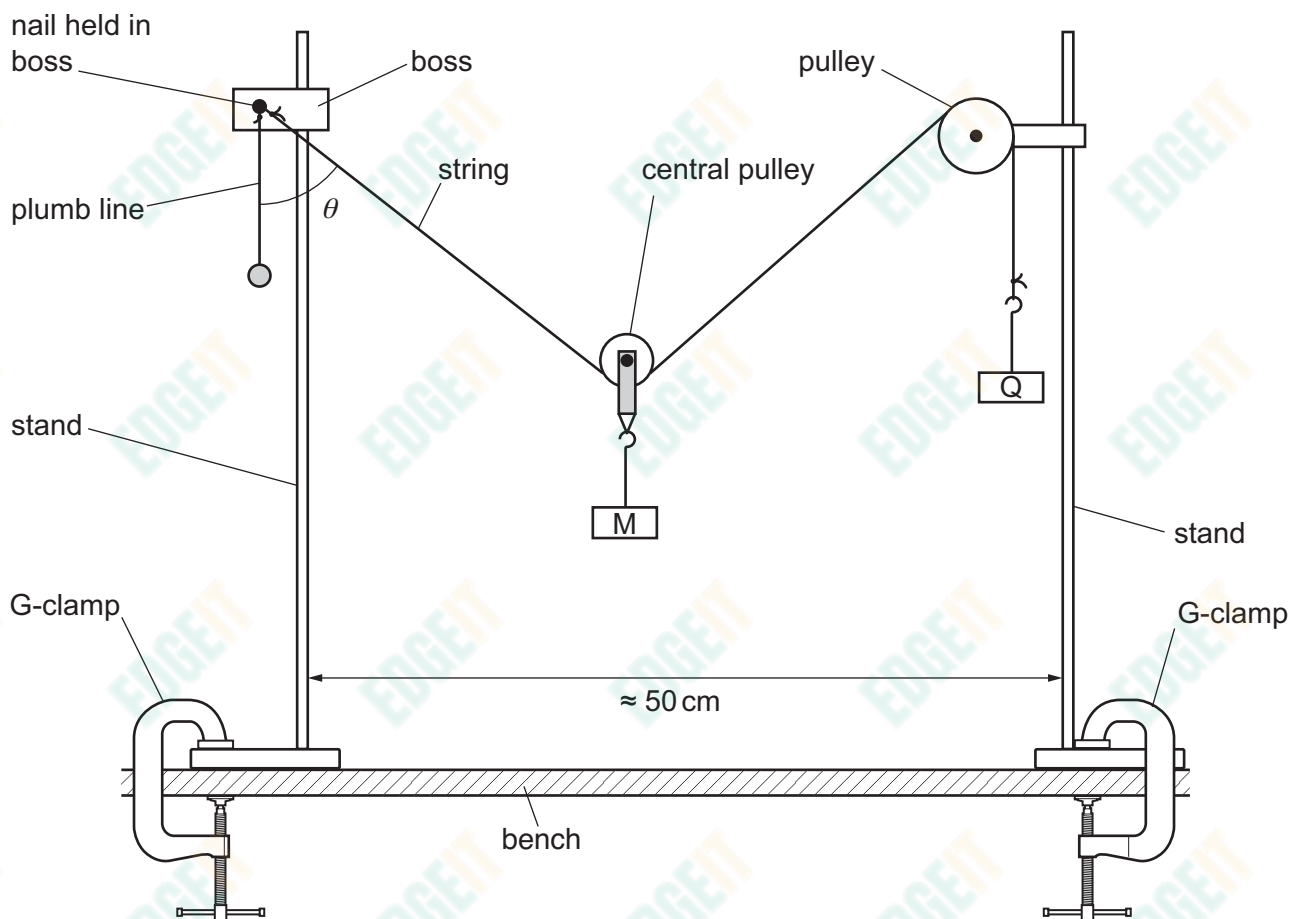


Fig. 1.1

- Adjust the pulley fixed to the stand so that the top of the pulley is approximately 60 cm above the bench.
- Adjust the boss holding the nail so that the nail is approximately 60 cm above the bench.





- Use some of the slotted masses to add a mass of 70 g to Q.
- The mass added to Q is x . Record the value of x .

$x = \dots\dots\dots$ g

- The angle between the plumb line and the string is θ , as shown in Fig. 1.1.

Measure and record θ .

$\theta = \dots\dots\dots^\circ$

- Carefully remove the slotted masses from Q.

[1]





(b) By using different numbers of slotted masses, vary x . For each value of x , measure θ .

Repeat until you have six sets of values of x and θ .

Record your results in a table. Include values of $\frac{1}{\cos \theta}$ in your table.

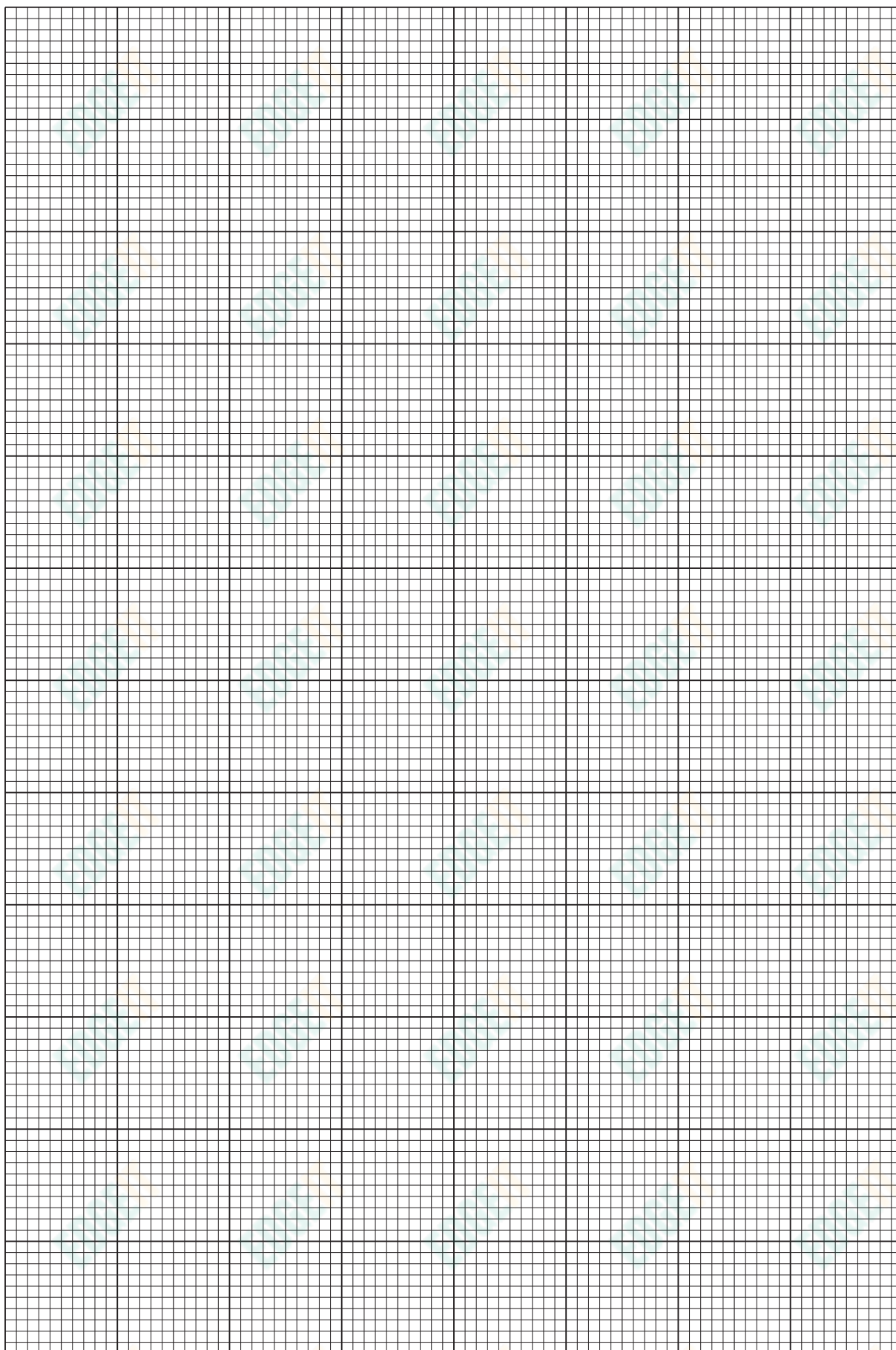
- (c) (i) Plot a graph of $\frac{1}{\cos \theta}$ on the y -axis against x on the x -axis. [10]
- (ii) Draw the straight line of best fit. [3]
- (iii) Determine the gradient and y -intercept of this line. [1]

gradient =

y -intercept =

[2]







- (d) It is suggested that the quantities θ and x are related by the equation

$$\frac{1}{\cos \theta} = ax + b$$

where a and b are constants.

Use your answers in (c)(iii) to determine the values of a and b .
Give appropriate units.

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots$$

[2]

- (e) The mass of M is M and the mass of Q is Q .

The constants a and b are related to M and Q by

$$a = \frac{2}{M} \quad \text{and} \quad b = \frac{2Q}{M}.$$

Calculate values for M and Q .

$$M = \dots\dots\dots \text{ g}$$

$$Q = \dots\dots\dots \text{ g}$$

[1]

[Total: 20]



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You may not need to use all of the materials provided.

2 In this experiment, you will investigate the effect of air resistance on a spinning card.

- (a) (i) • Assemble the apparatus as shown in Fig. 2.1, with the washer and plastic tube over the nail.

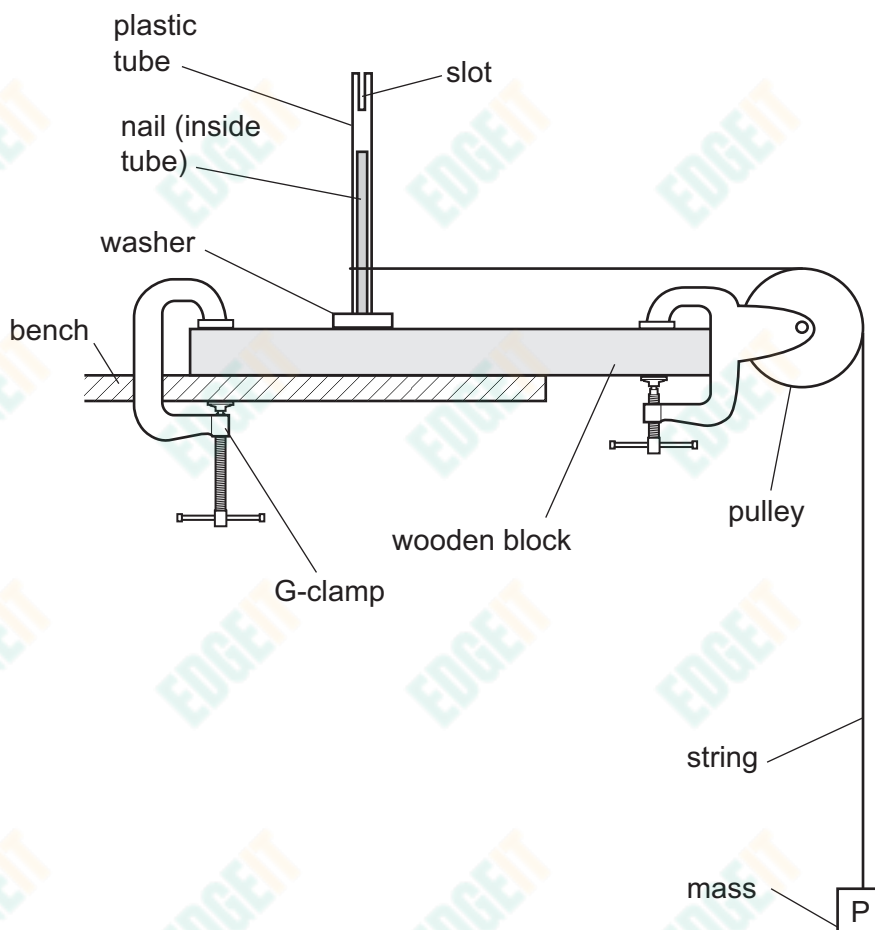


Fig. 2.1

- Rotate the plastic tube so that P rises until it just touches the pulley.
- Release the plastic tube so that P falls.
- Measure and record the time T_0 for P to reach the end of its fall.

$T_0 = \dots\dots\dots$ [2]



(ii) Estimate the percentage uncertainty in your value of T_0 . Show your working.

percentage uncertainty = % [1]

(b) (i) Fig. 2.2 shows card A.



Fig. 2.2

Measure and record the length L and the width W of the card, as shown in Fig. 2.2.

$L =$

$W =$ [1]

- (ii)
- Insert the card centrally into the slot at the top of the plastic tube with its length horizontal. If necessary, use a small piece of adhesive putty to fix it securely in the slot.
 - Rotate the plastic tube so that P rises until it just touches the pulley.
 - Release the plastic tube and measure the time T for P to reach the end of its fall.

$T =$ [2]





(c) Repeat (b) using the card marked B.

$L =$

$W =$

$T =$

[3]

(d) It is suggested that the relationship between T , T_0 , L and W is

$$k(T - T_0) = L^2 W$$

where k is a constant.

(i) Using your data, calculate two values of k .

first value of $k =$

second value of $k =$

[1]

(ii) Justify the number of significant figures that you have given for your values of k .

.....

.....

..... [1]





(e) It is suggested that the percentage uncertainty in the values of k is 15%.

Using this uncertainty, explain whether your results support the relationship in (d).

.....

.....

.....

..... [1]





- (f) (i) Describe **four** sources of uncertainty or limitations of the procedure for this experiment.

For any uncertainties in measurement that you describe, you should state the quantity being measured and a reason for the uncertainty.

1

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2

.....

3

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4

.....

[4]

- (ii) Describe **four** improvements that could be made to this experiment. You may suggest the use of other apparatus or different procedures.

1

.....

2

.....

3

.....

4

.....

[4]

[Total: 20]

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